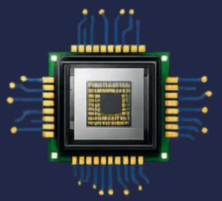
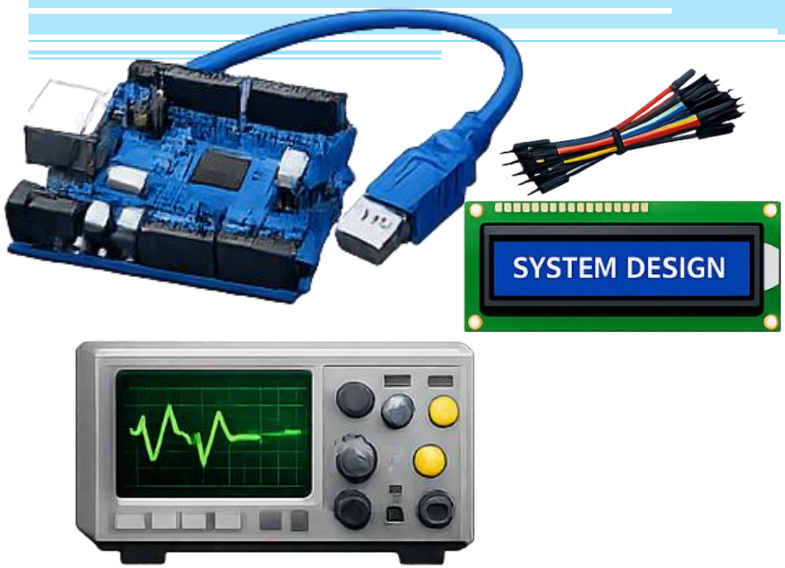
CS-301 Microprocessor-Based System Design



Lecture 13: ADCs and DACs

Course Teacher: Syeda Ramish Fatima



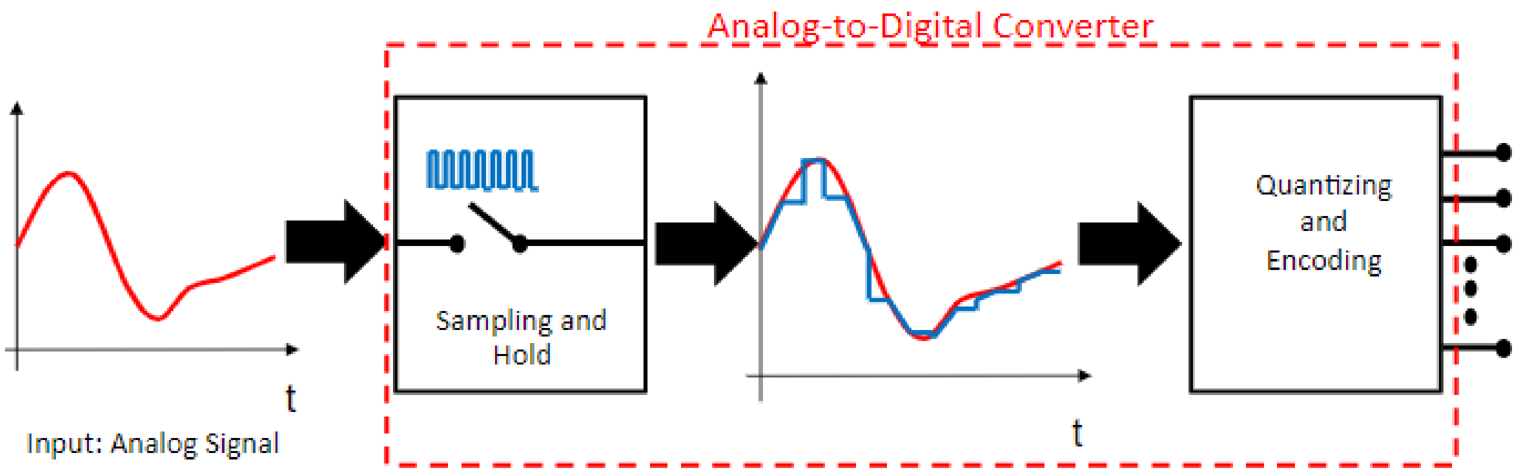
Analog to Digital Converter

- An electronic integrated circuit that converts an analog signal, such as a sound picked up by a microphone or light entering a digital camera, into a digital signal.
- Provides a link between the analog world of transducers and the digital world of signal processing and data handling.

ADC Conversion Process

Two main steps

- Sampling and holding
- Quantization and Encoding

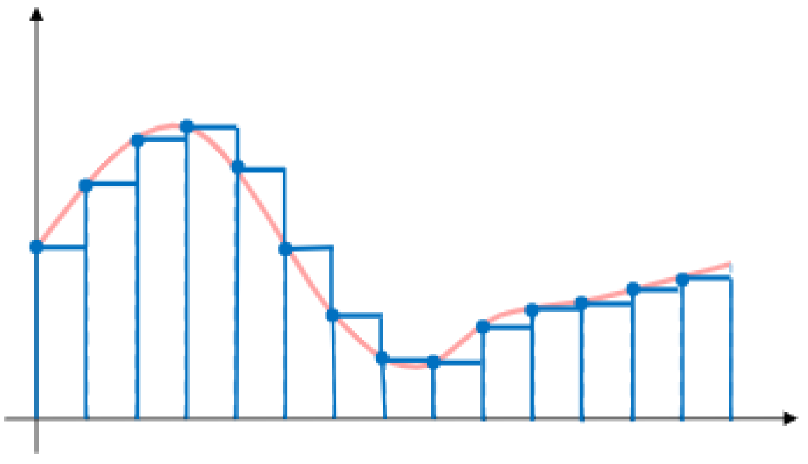


ADC Process

Sampling and Hold

- Measuring analog signal at regular time intervals.
- Sampling rate: the rate at which new digital values are sampled from the analog signal.
- Nyquist-Shannon sampling theorem => a faithful reproduction of the original signal is only possible if the sampling rate is higher than twice the highest frequency of the signal.

$f_{sample}>2*f_{signal}$



ADC Process

Sampling and Hold

- Since a practical ADC cannot make an instantaneous conversion, the input value must necessarily be held constant during the time that the converter performs a conversion (called the conversion time).
- The circuit is therefore named ***"sample and hold circuit"***

the conversion process take few second to yield results hence we need to hold the input vale before sampling A sample and hold circuit does exactly what the name says:

Sample → capture the signal's value at a specific instant
Hold → keep that value constant while the ADC converts it

1.23 → 1.2
2.87 → 2.8
3.01 → 3.0

rounding off means quantization

ADC Process

If N = 3 bits:

Possible binary outputs:

000
001
010
011
100
101
110
111

👉 Total = 8 levels (2^3)
K=8 quantization levels
0–1 V → 000
1–2 V → 001
...
7–8 V → 111
each level represnt the set of values

Quantization

More bits = more levels = better accuracy

- Quantizing the input samples on K discrete values

Q is the voltage gap between two adjacent levels

$$K = 2^N$$

- N is the number of bits of ADC
- Analog Quantization Size

$$Q = (V_{max}-V_{min})/2^N$$

- N is **resolution** in bits, Q is resolution in terms of voltage explained in the following slides.

ADC Process

Resolution

- it indicates the number of different, i.e. discrete, values it can produce over the allowed range of analog input values.
- a particular resolution determines the magnitude of the quantization error
- The input samples are usually stored electronically in binary form within the ADC, so the resolution is usually expressed as the bit depth.

Resolution

- For example, an ADC with a resolution of 8 bits can encode an analog input to one in 256 different levels ($2^8 = 256$).
- The values can represent the ranges from 0 to 255 (i.e. as unsigned integers) or from −128 to 127 (i.e. as signed integer), depending on the application.

Resolution

- Resolution can also be defined electrically, and expressed in volts.
- The change in voltage required to guarantee a change in the output code level is called the least significant bit (LSB) voltage.
- The resolution Q of the ADC is equal to the LSB voltage.

because in the difference of Vmax and Vmin the LSB is what only changed

Resolution

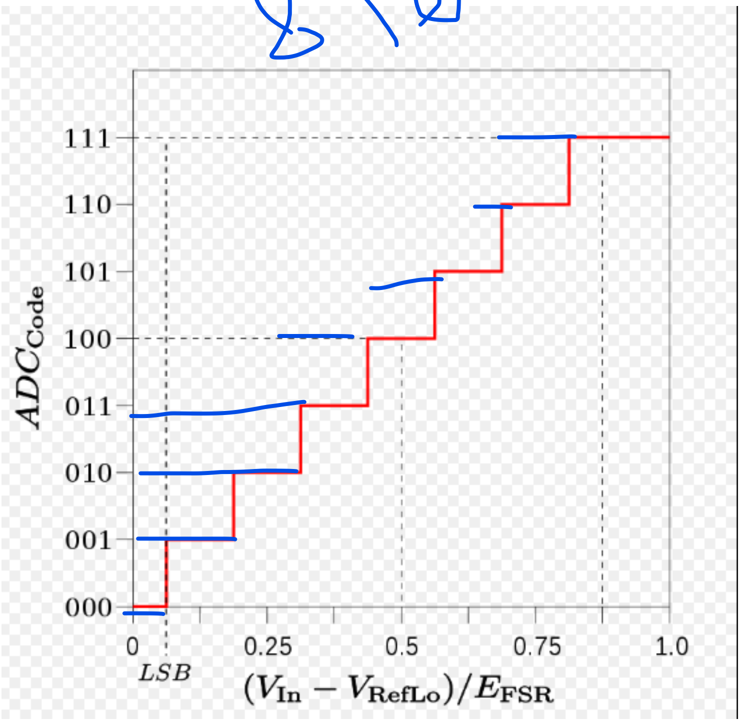
Example:

- Coding scheme as in figure
- Full scale measurement range = 0 to 1V
- ADC resolution is 3 bits => 8 quantization levels (codes)
- ADC voltage resolution,

1-0 = 1 $\quad Q = 1\ V / 8 = 0.125\ V.$

- In many cases, the useful resolution of a converter is limited by the signal-to-noise ratio (SNR) and other errors in the overall system expressed as an ENOB.

8 level



Resolution: ENOB

- Effective Number Of Bits
- It is a measure of the dynamic range of an analog-to-digital converter (ADC), digital-to-analog converter, or their associated circuitry.
- The resolution of an ADC is specified by the number of bits used to represent the analog value. Ideally, a 12-bit ADC will have an effective number of bits of almost 12.
- However, real signals have noise, and real circuits are imperfect and introduce additional noise and distortion. Those imperfections reduce the number of bits of accuracy in the ADC.
- The ENOB describes the effective resolution of the system in bits.
- An ADC may have a 12-bit resolution, but the effective number of bits, when used in a system, may be 9.

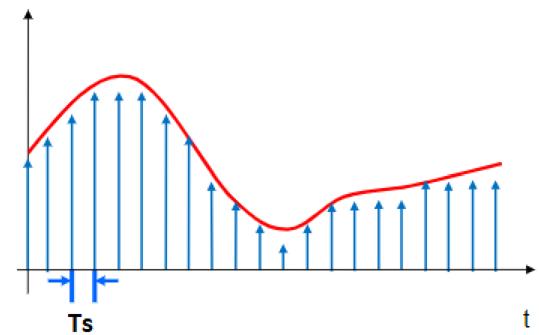
ENOB tells you how many bits of an ADC are actually “useful” after real-world imperfections like noise and distortion.

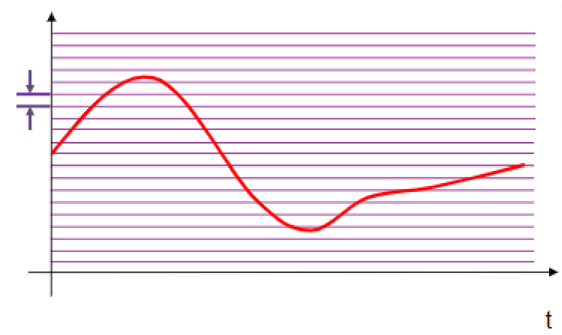
Even if an ADC is labeled “12-bit” or “16-bit,” it may not perform perfectly at that level. ENOB gives the true effective resolution.

ADC Process Accurracy

The accuracy can be improved by increasing

- Sampling rate Ts
  - Increases the maximum frequency that can be measured
- Resolution (bit depth)
  - Improves accuracy in measuring the amplitude of analog signal





ADC Errors

- **Aliasing**
- Occurs when the input signal is changing must faster than the sampling rate
- Should follow the Nyquist-
- Shannon Theorem
- $f_{sample} > 2 * f_{signal}$

Aliasing happens when:
A signal is sampled too slowly, so it gets misrepresented as a lower-frequency signal.
In simple words:
The sampled signal “lies” about how fast it was actually changing.

- **Quantization Error**
- The inherent uncertainty in digitizing an analog value as a result of the finite resolution of the conversion process.
- It depends on the number of bits in the converter, along with its errors, noise, and nonlinearities

when we round off, we loose precision that is quantization error

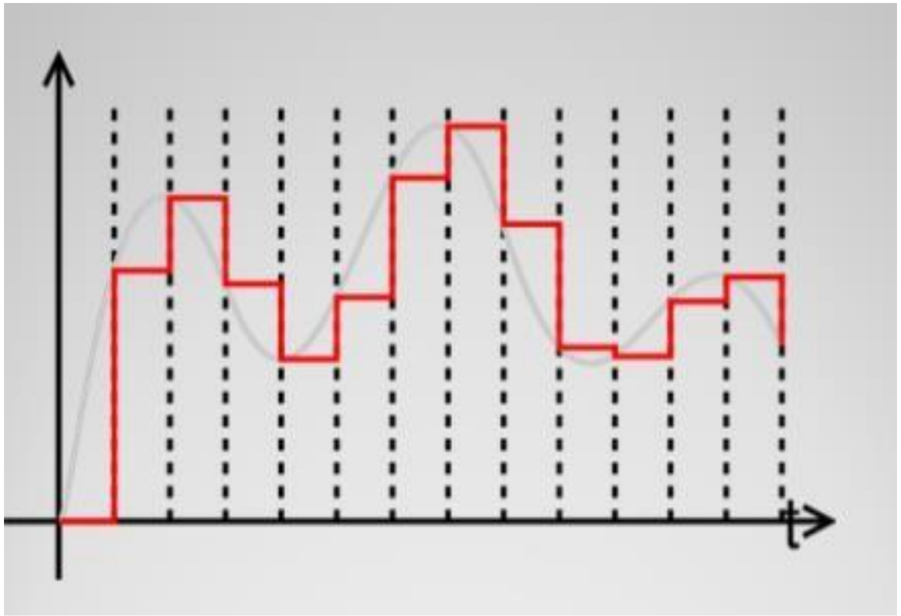
Digital to Analog Converter (DAC)

Digital to Analog Converter

- It is a system that converts a digital signal into an analog signal.
- It converts an abstract finite-precision number (usually a fixed-point binary number) into a physical quantity (e.g., a voltage or a pressure).
- In particular, DACs are often used to convert finite-precision time series data to a continually varying physical signal.

DAC

- Generates piecewise continuous signal from digital code.



DAC

There are 6 main specifications:

1. Reference Voltage
2. Resolution
3. Linearity
4. Monotonicity
5. Speed
6. Settling Time
7. Error

DAC Specifications

Reference Voltage

To a large extent the output properties are determined by the reference voltage.

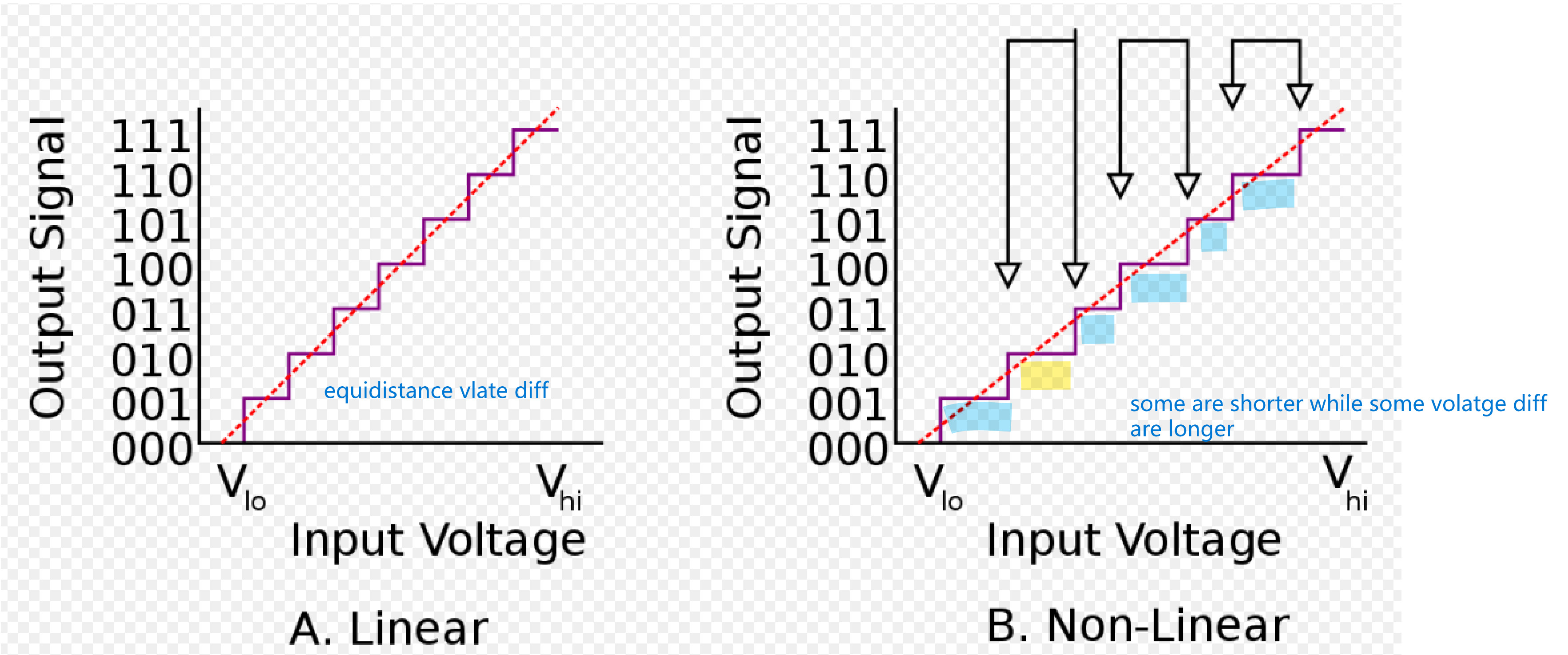
- ❖ Non-Multiplier DAC – the reference voltage is constant and is set by the manufacturer
- ❖ Multiplier DAC – the reference voltage can be changed during operation

A reference voltage is the stable voltage that an ADC uses as its “scale” for converting analog values into digital numbers.

DAC Specifications

Linearity

Relationship between the output voltage and the digital signal input.

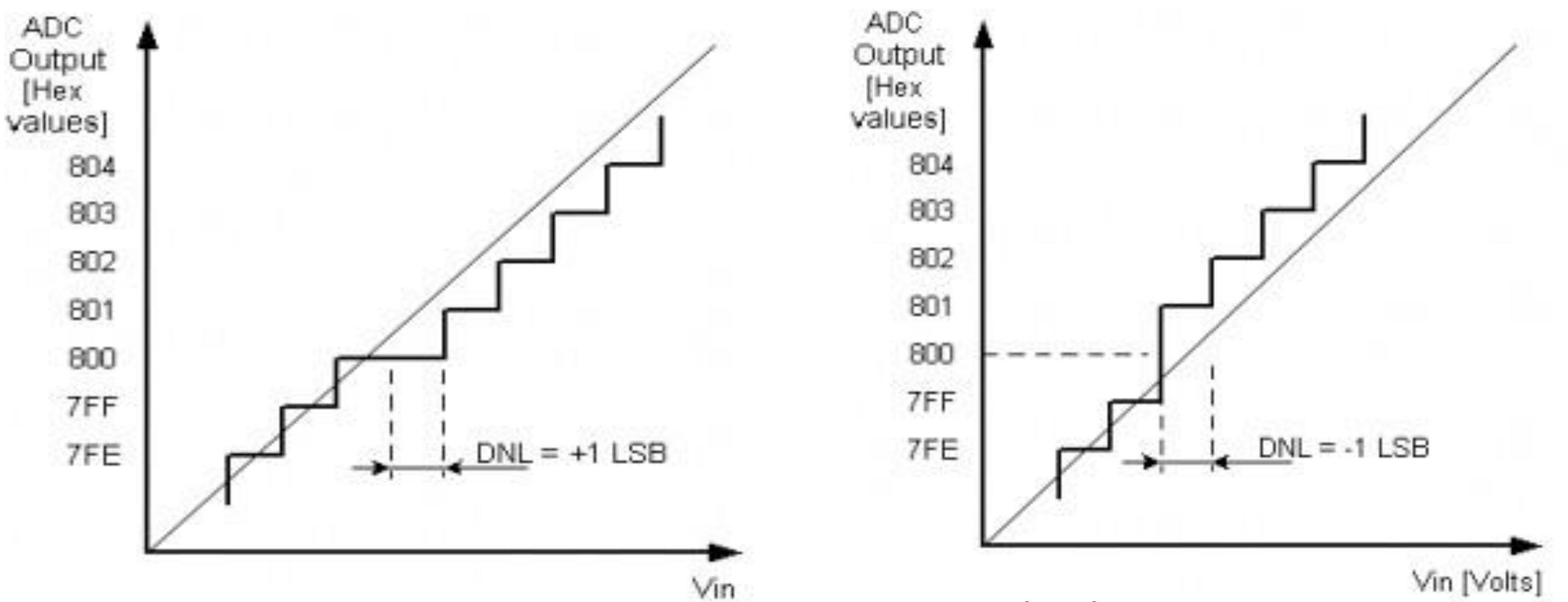


DAC Specifications - Linearity

Differential nonlinearity (acronym DNL)

- It is a commonly used measure of performance in digital-to-analog (DAC) and analog-to-digital (ADC) converters.
- It is a term describing the deviation between two analog values corresponding to adjacent input digital values.
- It is also known as a missing code.

DAC Specifications - Linearity



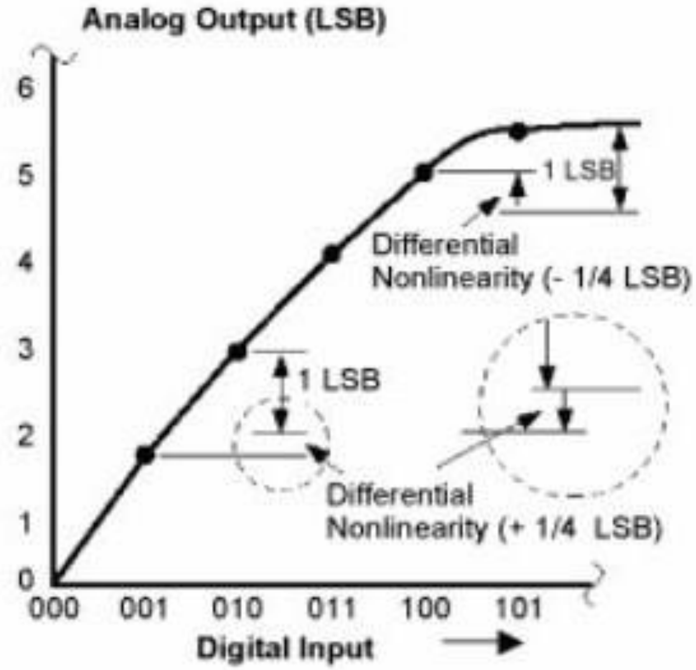
Differential Non-Linearity (DNL)

normally there was always 1V differnce but now at specific points we have +1 and -1 deviation

DAC Specifications - DNL

- Can also be defined as the difference between an actual step height and the ideal value of 1LSB.
- Should be less than or equal to one to ensure monotonicity.

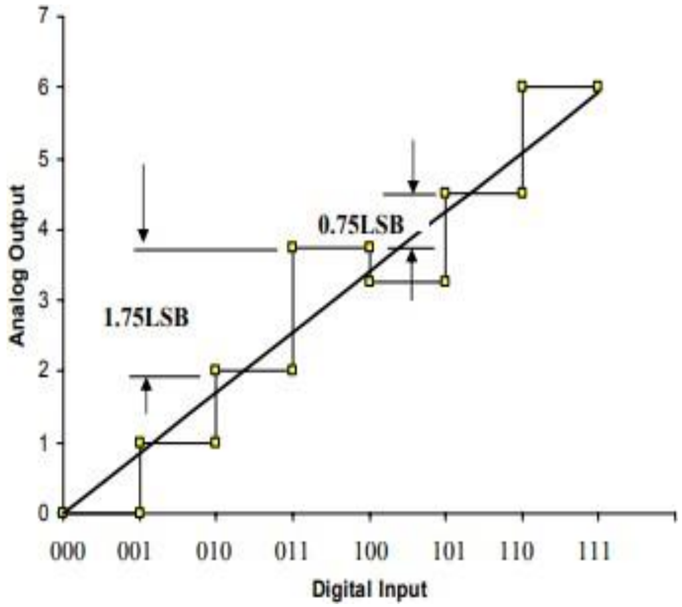
As the input increases, the output never decreases.



DAC Specifications

Monotonicity

- A monotonic DAC yields an increase in output as input increases.
- If a differential non-linearity of greater than 1LSB occurs, increasing the digital input may actually result in a decreased output.



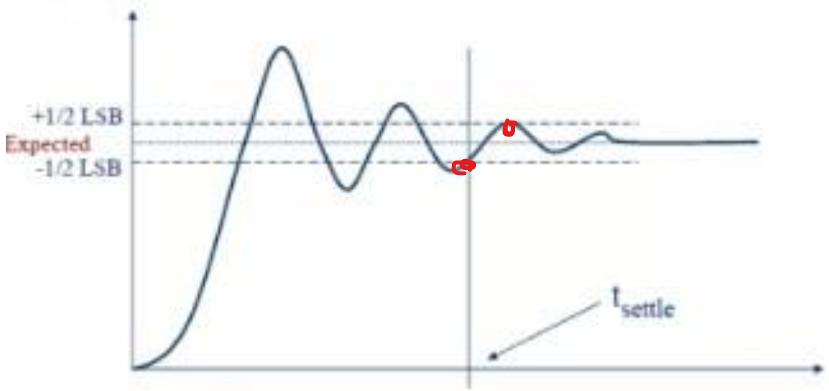
DAC Specifications - Speed

- It is the rate at which the input register is updated.
- $DAC_{\text{High speed}} \approx 1$ MHz
- Some state-of-the-art 12-16 bit DACs may reach a speed of 1GHz.
- Speed is limited by the clock speed and settling time of the DAC.

DAC Specifications

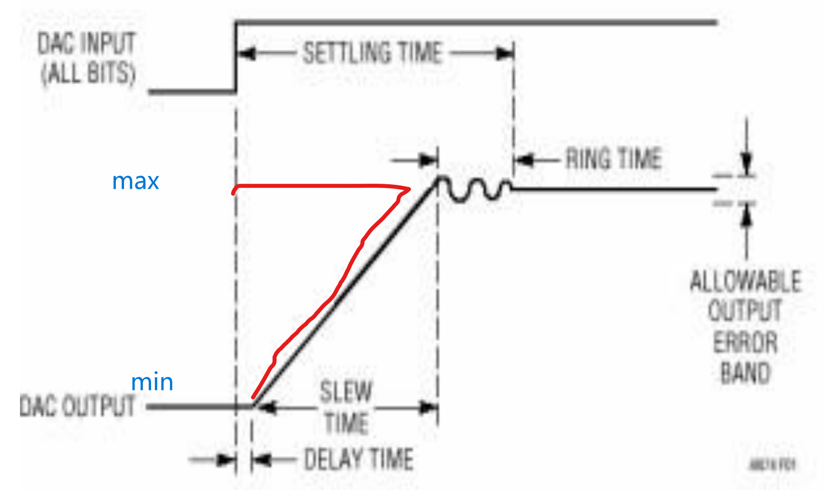
Settling Time

- Ideally, an instantaneous change in analog voltage would occur when a new binary word enters into a DAC.
- Settling time is the time taken by the DAC to reach $\pm$½ of the LSB of its new voltage.



DAC Specifications – Settling Time

- Components include delay, slew time, and ring time.
- Fast converters reduce slew time, but usually result in longer ring times.
- Delay time is normally a small term.



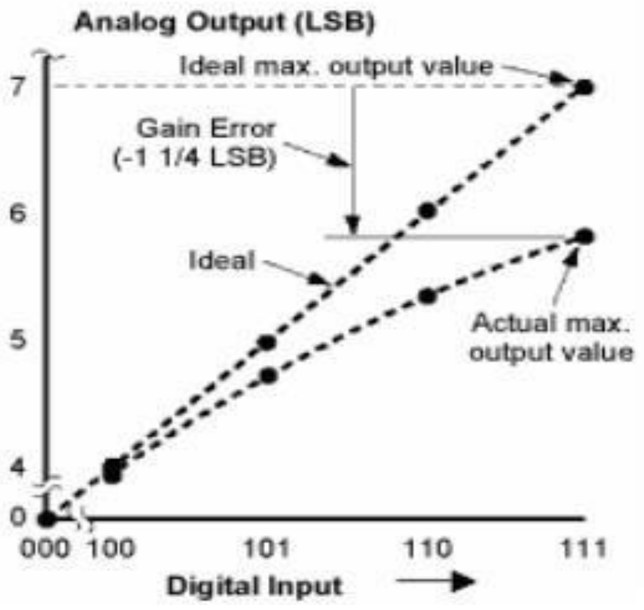
Delay is the time taken for an input change to appear at the output.

Slew time is the time it takes for a signal to change from one level to another due to limited speed of the circuit.

DAC Specifications - Error

Gain Error

- Defined as the difference between the ideal max output voltage and the actual max output voltage (after subtracting offset error).
- Changes the slope of the output, thereby creating the same percentage error for each step.
- Expressed in mV as a percentage of the maximum output.



DAC Specifications - Error

Offset Error

- The offset error equals the analog output when the digital signal is zero.
- Typically defined in absolute millivolts with (10mV being acceptable).